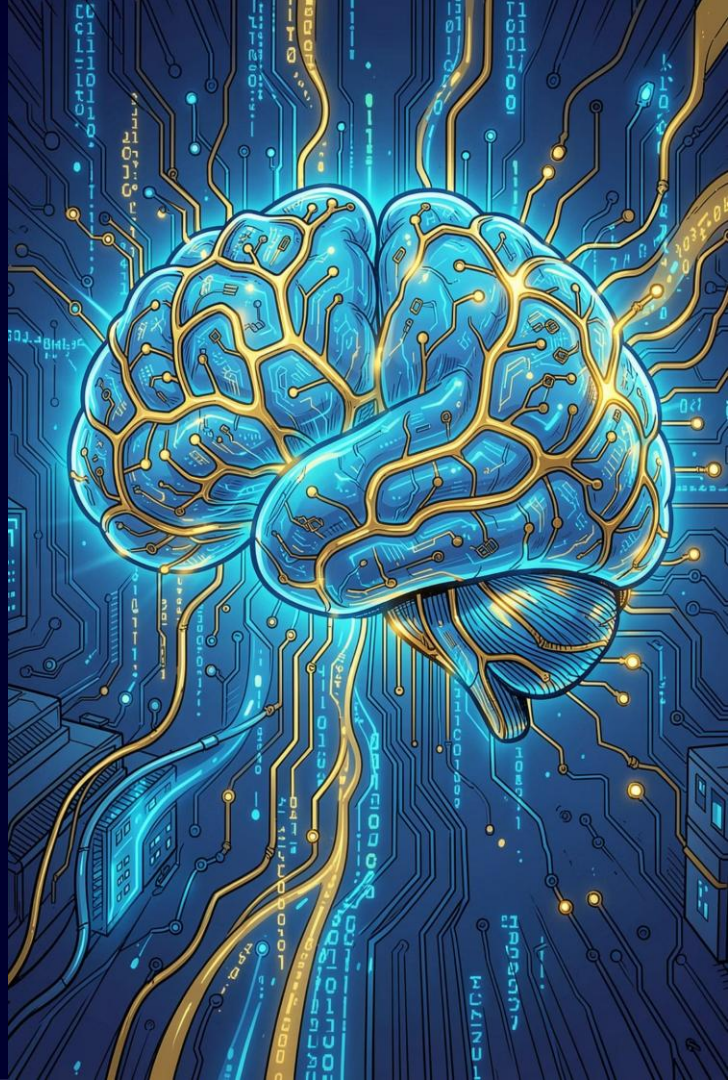
Introduction to Artificial Intelligence

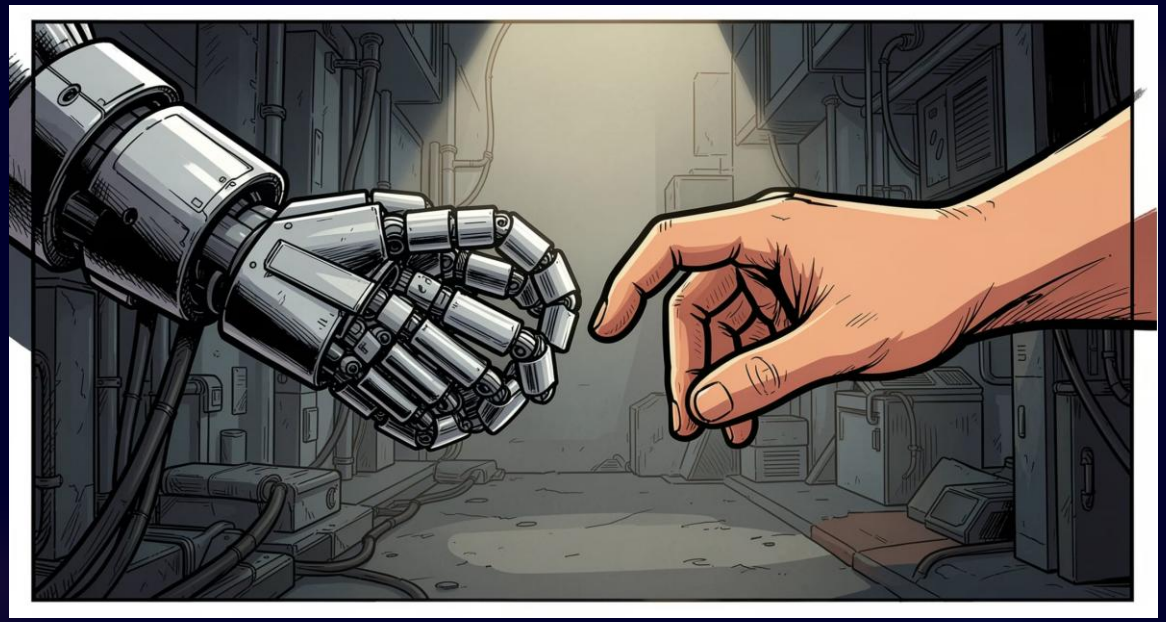
A comprehensive overview of AI's structure, history, definitions, approaches, and current capabilities — from philosophical foundations to modern intelligent systems.

AI FUNDAMENTALS

COMPUTER SCIENCE

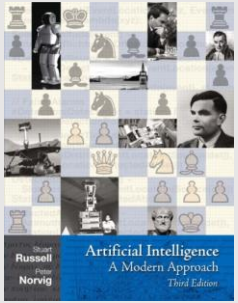


What Is AI?



Textbooks

- ****** Russell & Norvig, AI: A Modern Approach, 3rd Edition**
- **Artificial Intelligence, Kevin Knight, Elaine Rich, B. Shivashankar Nair, 2nd Edition,**
- **2008**
- **Artificial Neural Networks, B. Yagna Narayana, PH**





INTRODUCTION TO

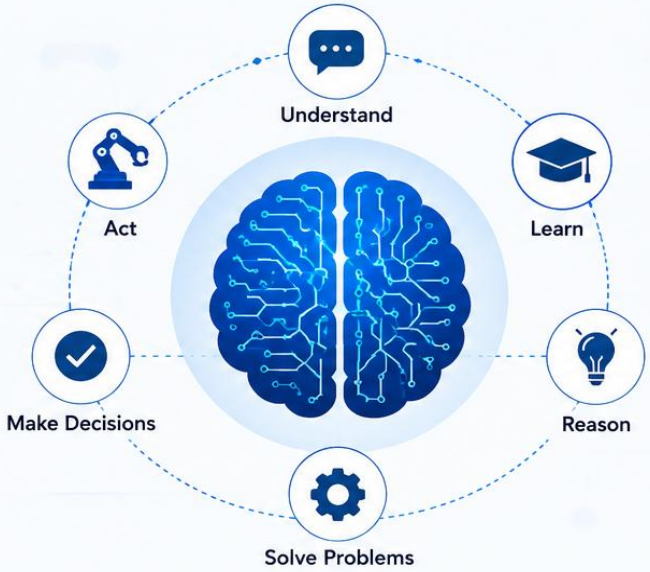
ARTIFICIAL INTELLIGENCE (AI)

UNIT – I OVERVIEW

1. WHAT IS ARTIFICIAL INTELLIGENCE?

Artificial Intelligence (AI) is the simulation of human intelligence in machines that are programmed to **think, learn, reason, and act like humans.**

It enables machines to perform tasks that typically require human intelligence, such as understanding language, recognizing patterns, solving problems, and making decisions.



In Simple Words: AI is about building smart machines that can think, learn, and act to solve problems and help humans in everyday life.

DEFINITION OF ARTIFICIAL INTELLIGENCE

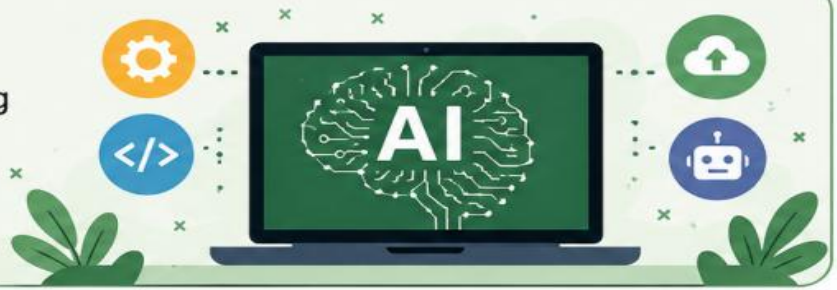
“ Artificial Intelligence is the branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, problem-solving, decision-making, and understanding language. ”



DEFINITION OF AI

“ Artificial Intelligence is the science and engineering of making intelligent machines, especially intelligent computer programs. ”

– John McCarthy, 1956





DEFINITION OF ARTIFICIAL INTELLIGENCE

“ Artificial Intelligence is the study of intelligent agents—systems that perceive their environment and take actions to maximize the chances of achieving their goals. ”

– Stuart Russell & Peter Norvig

DIFFERENT DEFINITIONS OF ARTIFICIAL INTELLIGENCE



1. John McCarthy (1956)

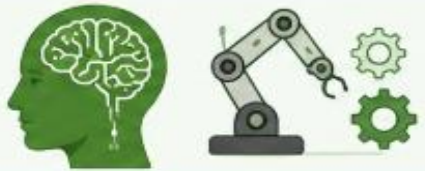
> "Artificial Intelligence is the science and engineering of making intelligent machines, especially intelligent computer programs."





2. Marvin Minsky

> "Artificial Intelligence is the science of making machines do things that would require intelligence if done by humans."





3. Elaine Rich & Kevin Knight

> "Artificial Intelligence is the study of how to make computers do things at which, at the moment, people are better."





4. Stuart Russell & Peter Norvig

> "Artificial Intelligence is the study of intelligent agents—systems that perceive their environment and take actions to maximize the chances of achieving their goals."



5. IBM

> "Artificial Intelligence is technology that enables computers and machines to simulate human intelligence and problem-solving capabilities."



Defining Artificial Intelligence

Scholars have offered many definitions of AI over the decades. Collectively, they reveal four fundamental perspectives on what it means for a system to be "intelligent."

> "The branch of computer science that is concerned with the automation of intelligent behavior." — **Luger and Stubblefield, 1993**

> "The study of the computations that make it possible to perceive, reason, and act." — **Winston, 1992**

> "The art of creating machines that perform functions that require intelligence when performed by people." — **Kurzweil, 1990**

> "The study of how to make computers do things at which, at the moment, people are better." — **Rich and Knight, 1991**

> "A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes." — **Schalkoff, 1990**

> "The exciting new effort to make computers think — machines with minds, in the full and literal sense." — **Haugeland, 1985**

These definitions lead to four possible conclusions about AI: systems that **think like humans**, systems that **think rationally**, systems that **act like humans**, and systems that **act rationally**. The textbook advocates **"acting rationally"** as the most productive framework.

Made with GAMMA

Four Approaches to AI

Think Like Humans

Cognitive modeling of brain processes

Think Rationally

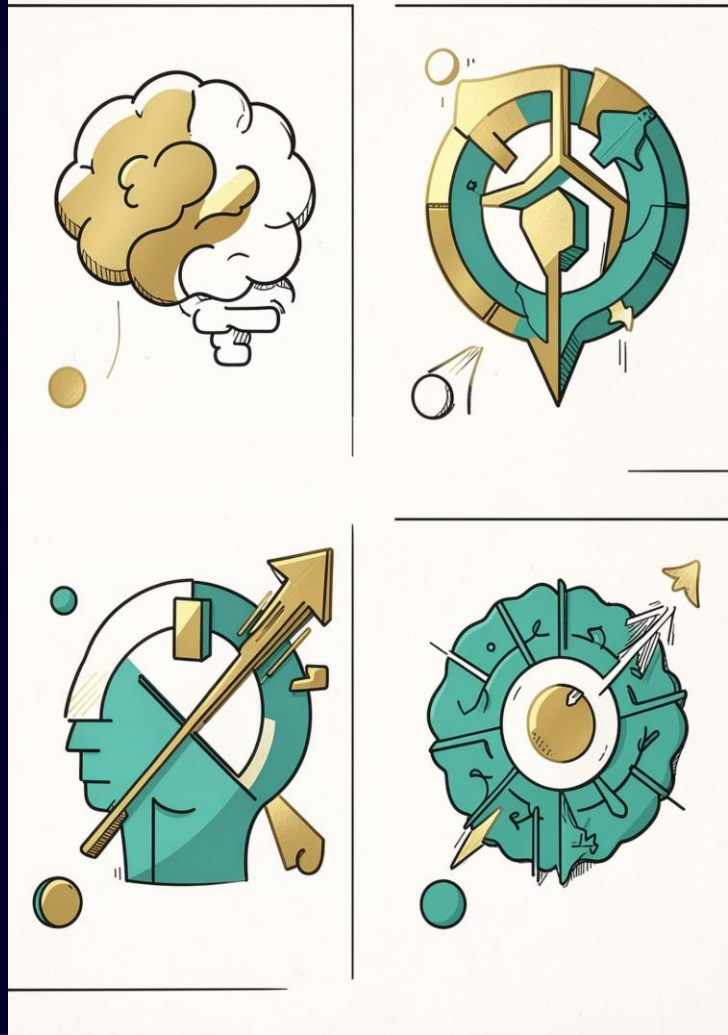
Laws of logic and correct reasoning

Act Like Humans

Turing Test — imitation game

Act Rationally

Maximize goal achievement — **textbook's preferred approach**



Four Approaches to Defining AI



Think Like Humans

Cognitive modeling — requires scientific theories of internal brain activity. Validation via behavioral prediction or neurological data. Now distinct from AI.



Think Rationally

"Laws of thought" — Aristotle's logic, formal derivation. Problem: not all intelligent behavior is logical deliberation.



Act Like Humans

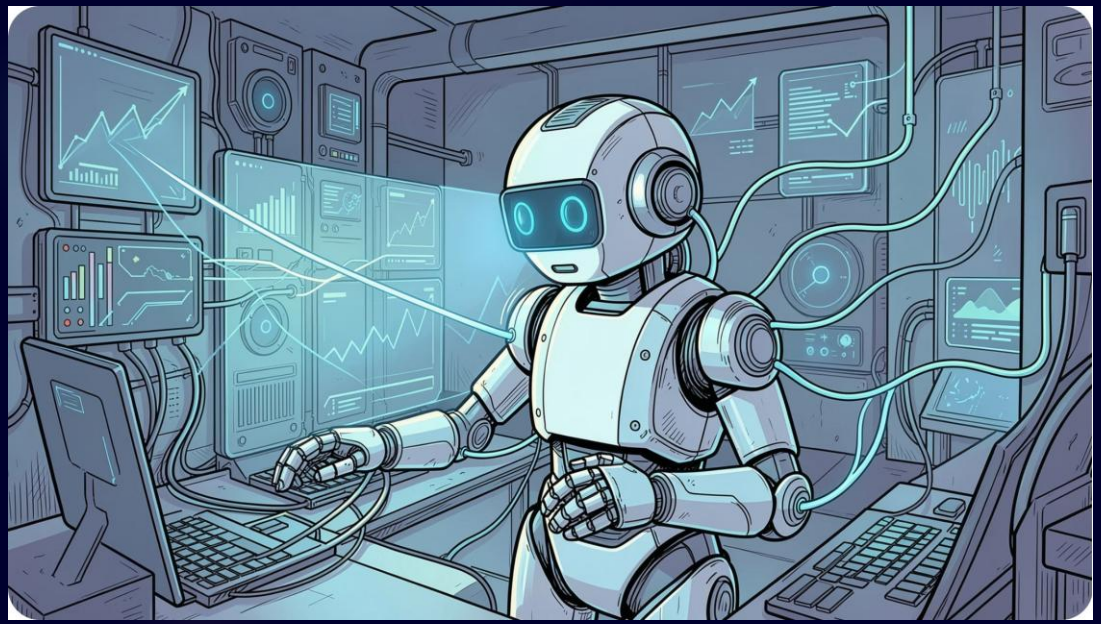
Turing Test (1950) — imitation game. Requires knowledge, reasoning, language understanding, and learning. Still relevant today.



Act Rationally

Rational agent: does the right thing — maximizing goal achievement given available information. **Textbook advocates this approach.**

Rational Agents & Key Challenges



Rational Agent

An **agent** perceives and acts. Abstractly, a function from percept histories to actions:

$$f : P \rightarrow A$$

We seek the agent with best performance for a given environment. Computational limits make *perfect* rationality unachievable.

Big Obstacles

- **rain knowledge gap** — too little understanding of how context is inferred
- **Prior knowledge** — machines need environmental knowledge to match human behavior
- **rain models may be counter-productive** — attack problems given the nature of digital computers

GOALS OF AI

01

CREATE INTELLIGENT SYSTEMS

To create intelligent systems that can perform tasks efficiently and accurately.



02

ENABLE LEARNING AND ADAPTATION

To enable machines to learn from experience and adapt to new situations.



03

AUTOMATE COMPLEX PROCESSES

To automate complex processes and assist in decision making.



04

ENHANCE HUMAN CAPABILITIES

To enhance human capabilities and improve quality of life.



05

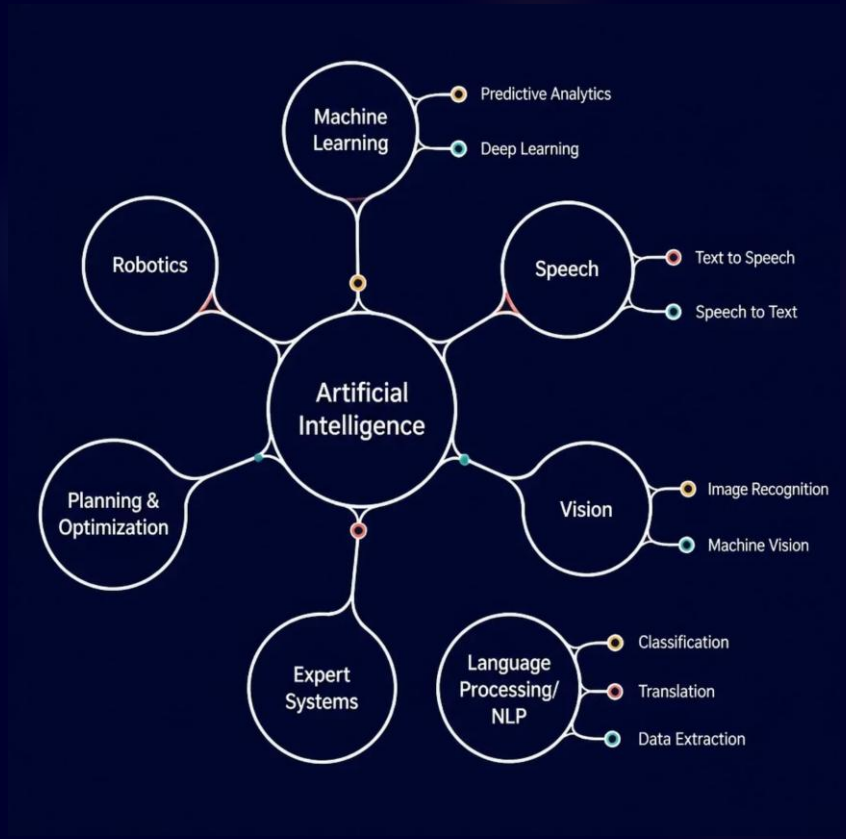
SOLVE REAL–WORLD PROBLEMS

To solve real–world problems in areas like health, education, environment, business and more.



In Simple Words: The goal of AI is to build smart machines that can learn, think, decide, and help solve problems for a better tomorrow.

AI Branches at a Glance



Seven Core Domains

AI spans seven major fields — from **Machine Learning** and **NLP** to **Robotics** and **Vision** — each with specialized sub-fields driving real-world applications.

Structure of Artificial Intelligence

Artificial Intelligence encompasses a broad family of subfields, each addressing a different aspect of intelligent behavior. At its core, AI branches into seven major domains, many of which contain specialized sub-techniques.

Machine Learning

- Predictive Analytics
- Deep Learning

Speech

- Text to Speech
- Speech to Text

Vision

- Image Recognition
- Machine Vision

NLP

- Classification
- Translation
- Data Extraction

Expert Systems

Knowledge-based reasoning systems

Planning & Optimization

Goal-directed decision making

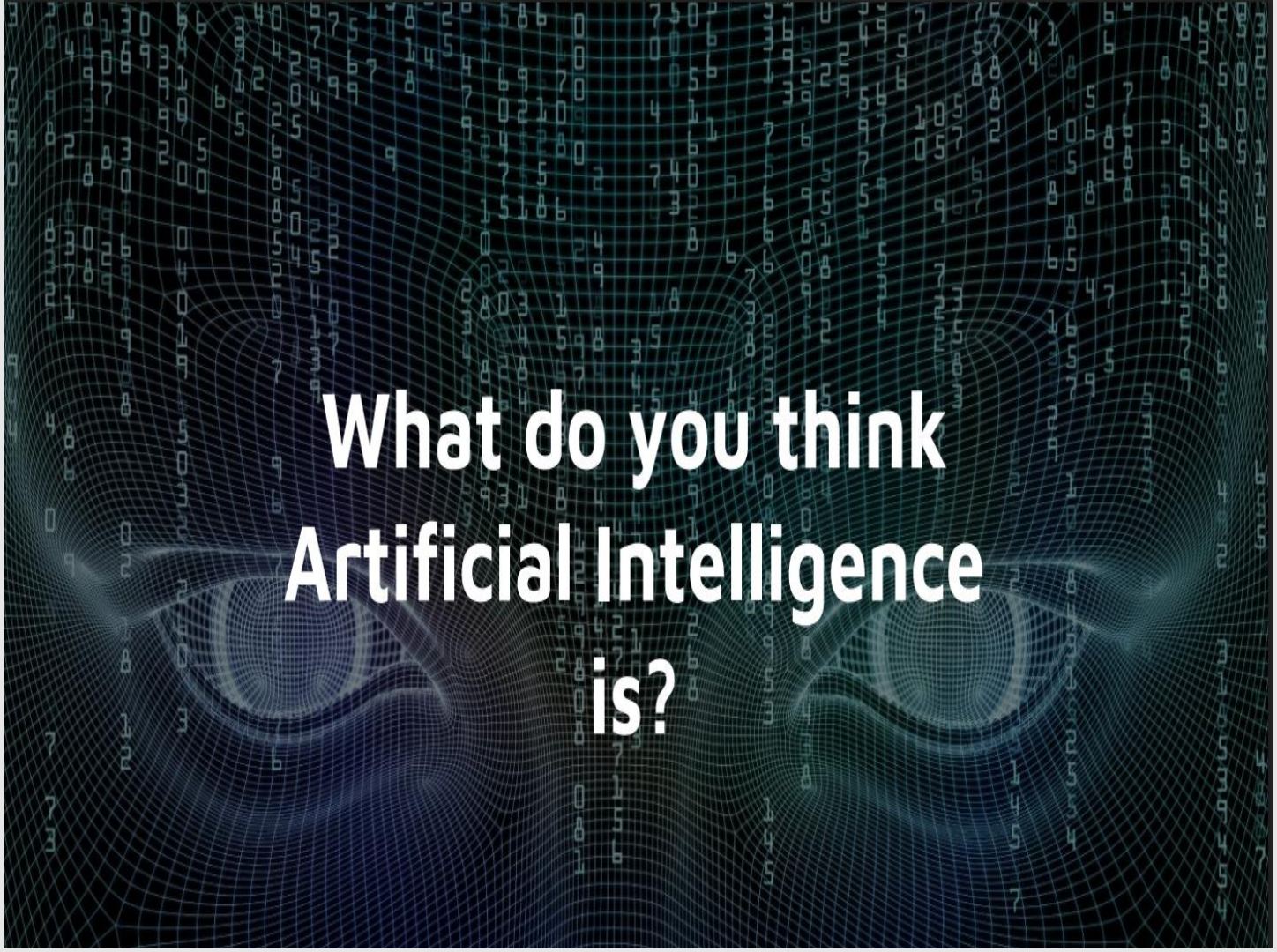
Robotics

Physical intelligent agents

These domains collectively define the landscape of AI research and application, from perceiving the world (vision, speech) to reasoning about it (planning, expert systems) and acting within it (robotics).

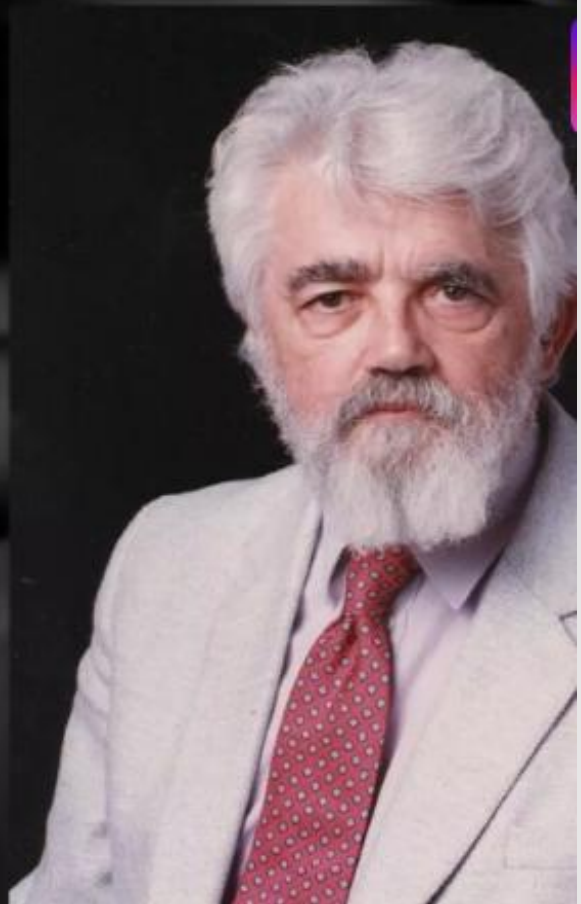

ARTIFICIAL INTELLIGENCE


AI
0101
0101
0101


What do you think
Artificial Intelligence
is?

FATHER OF AI

JOHN MACCARTHY WHO COINED THE TERM ARTIFICIAL INTELLIGENCE



Military Technology



Aiming, Defending, and Moving are done simultaneously by us.



But with AI technology, the ank will be able to auto-detect enemy units and take action




Space Exploration


Underwater Mining


Medical Research



MANAV

is India's first 3D printed expandable platform robot which was developed in the laboratory of A-SET Training and Research Institutes by Diwakar Vaish(head Robotics and Research.



SOFIA

THE HUMONOID ROBOT

SOFIA THE ROBOT

- Sophia's creator, Dr. David Hanson, is the founder of Hanson Robotics . After working at Disney ,Dr. Hanson aspired to create genius machines that will surpass human intelligence

- On November 21, 2017 Sophia was named the United Nations Development Programme first ever Innovation Champion for Asia and the Pacific.

- Sophia is an evolving genius machine. Her incredible human likeness, expressiveness.

- Sophia is a highly sought-after speaker in business and showed her power and great potential across many industries.
- She has met face-to-face with key decision makers in banking, insurance, auto manufacturing, property development, media, and entertainment.



RASHMI ROBOT

NEWS

INDIA'S VERSION OF HUMANOID ROBOT SOPHIA TO SPEAK BHOJPURI, MARATHI



Rashmi Robot can speak Three languages as Hindi,Marathi,English.

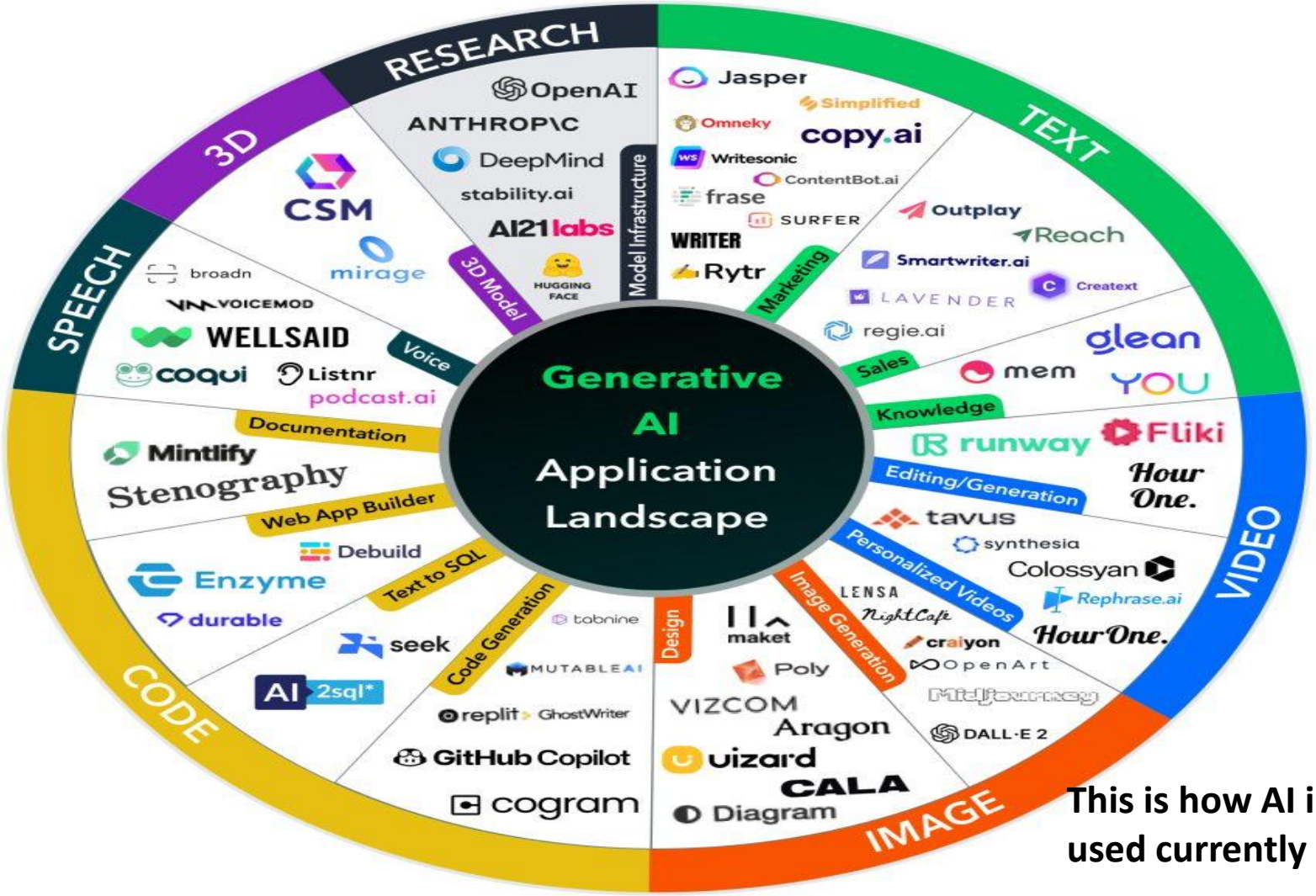
- Ranjith has developed the Rashmi in TWO years at the cost of 50000.
- It can talk for hours with a person and it recognizes the person after few meeting due to camera fitted in his eyes.

Rashmi uses linguistic interpretation (LI), artificial intelligence (AI),



Introduction to Intelligence Vs AI

INTELLIGENCE	ARTIFICIAL INTELLIGENCE
It is a natural process.	It is programmed by humans.
It is actually hereditary.	It is not hereditary.
Knowledge is required for intelligence.	KB and electricity are required to generate output.
No human is an expert. We may get better solutions from other humans.	Expert systems are made which aggregate many persons' experience and ideas.



This is how AI is being used currently

Turing Approach/Acting Humanly



Foundations of AI

Philosophy(428 B.C to present)

- How knowledge has come?
- What is knowledge?

- Through observation
- Through Confirmation
- Through use of awareness
- Through inducing and all etc.,…
- Dualism(Body &Soul)
- Materialism(body)

Mathematics(C.800 –present)

- What are the formal rules/conclusions?
- What is logic
- How is it being proved
- What are instances..
- How do we prove..
  Etc.,

- Through axioms/rules
- Through algorithms
- Through proofs
- Probability and Random functions

Economics(1776 –present)

- How should we make decisions?
- What is future rate of growth?

- Through Decision theory
- Games
- Operations research
- Survey
- General awareness

Neuroscience(1861 –present)

- How do brains process information?
- How do they/we remember?
- How is information

- Through Neuroscience
- Through cognition
- Through neurons
- Through sensors
- Through synapses

Psychology(1879-present)

- How do humans and animals think and learn?
- How do they act?
- How do they respond?

- Through Behaviorism
- Through sensors/Snesing
- Through cognitive psychology
- Through perceiving

Computer Engineering (1940 –present)

- How can we build an efficient computer?

- Through programming
- Through logic functions
- Through iterations
- Through knowledge setup
- Build an efficient computer (HW/SW)

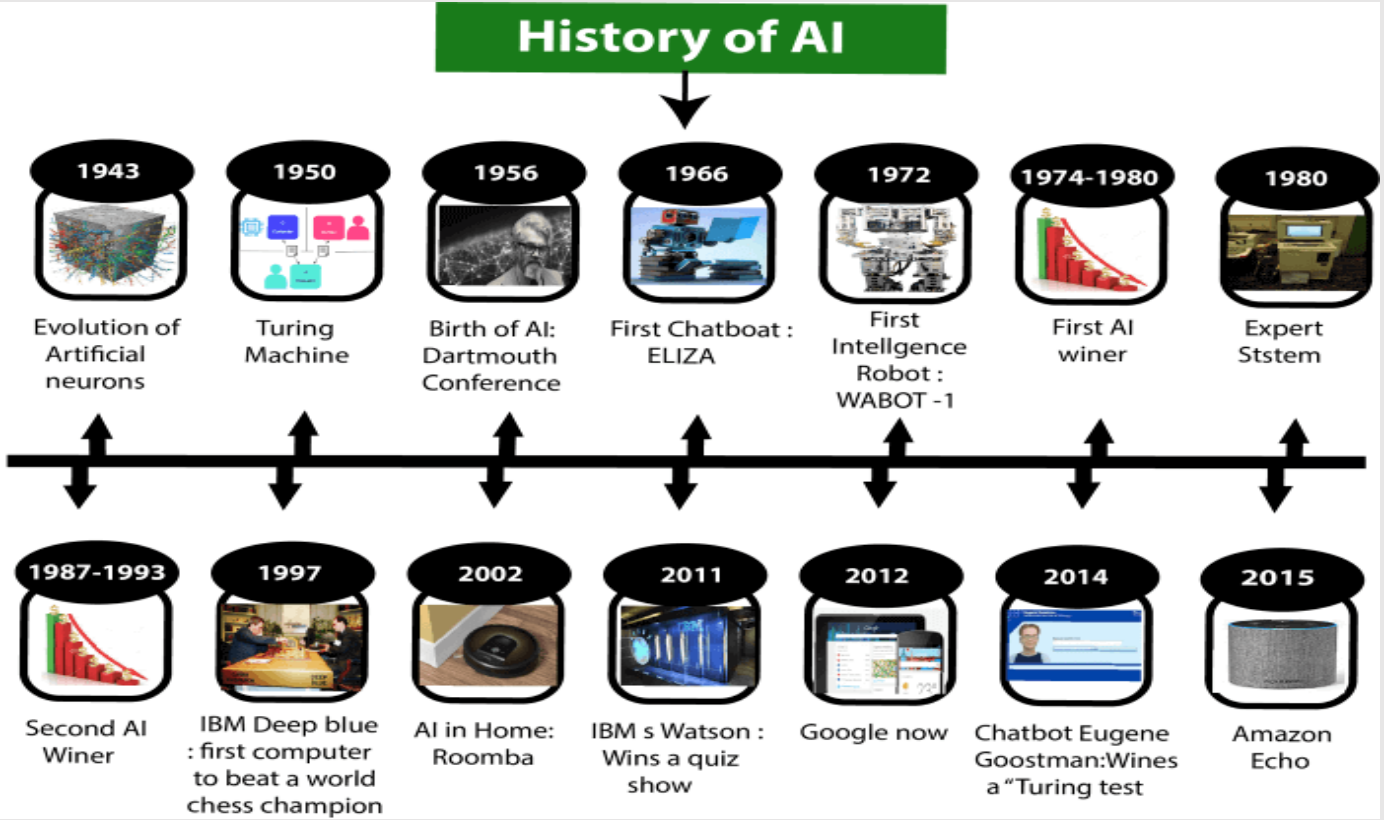
Control Theory and Cybernetics(1948-Present)

- How can artifacts operate under their own control?

- Through security
- Through objective function
- Through control theory

History of AI

- journey from the AI generation to till date development.



In general, And would find in most of the search results

History of AI :Different timelines, Different Researchers

History of AI

The emergence of intelligent agents (1993-2011)

- **Year 1997:** In the year 1997, **IBM Deep Blue** beats world chess champion, Gary Kasparov, and became the first computer to beat a world chess champion.
- **Year 2002:** for the first time, AI entered the home in the form of **Roomba**, a vacuum cleaner.
- **Year 2006:** AI came in the Business world till the year 2006. Companies like **Facebook, Twitter, and Netflix** also started using AI.

1997: Deep Blue ends Human Supremacy in Chess



Deep Blue had Kasparov in deep thought (CNN)

vs.



The cunning Deep Blue (CNN)

I could feel human-level intelligence across the room

-Gary Kasparov, World Chess Champion (human)

In a few years, even a single victory
in a long series of games would be the triumph of human genius.

NPTEL

State of Art of AI: Currently

- Autonomous Planning and scheduling
- Game Playing
- Autonomous control
- Diagnosis
- Logistic Planning
- Robotics
- Language understanding and Problem solving
- Recommendations
- Predictions and many more…**Now AI is a part of Ecosystem in itself…..**

Types of Artificial Intelligence:

Types of Artificial Intelligence

Weak AI or Narrow AI:

- Narrow AI perform a dedicated task with intelligence. Currently available AI is Narrow AI.
- Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task. Hence it is also termed as weak AI.
- **Apple Siri** is a good example of Narrow AI.
- **IBM's Watson** supercomputer also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing.
- Some Examples of Narrow AI are playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition.

General AI:

- General AI is a type of intelligence which could perform any intellectual task with efficiency like a human.
- The idea behind the general AI to make such a system which could be smarter and think like a human by its own.
- Currently, there is no such system exist which could come under general AI and can perform any task as perfect as a human.

Super AI:

- Super AI :machines could surpass human intelligence, and can perform any task better than human with cognitive properties.
- Machine can have ability to think, to reason , solve the puzzle, make judgments, plan, learn, and communicate by its own.
- Super AI is still a **hypothetical** concept of Artificial Intelligence.
- Development of such systems in real is still world changing task.

Artificial Intelligence type-2: Based on functionality

1. Reactive Machines

- Such AI systems do not store memories or past experiences for future actions.
- These machines only focus on current scenarios and react on it as per possible best action.
- **IBM's Deep Blue** system is an example of reactive machines.
- Google's AlphaGo is also an example of reactive machines.

Artificial Intelligence type-2: Based on functionality

2. Limited Memory

- Limited memory machines can store past experiences or some data for a short period of time.
- These machines can use stored data for a limited time period only.
- **Self-driving cars** are one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

Artificial Intelligence type-2: Based on functionality

3. Theory of Mind

- Theory of Mind AI should understand the human emotions, people beliefs, and be able to interact socially like humans.
- This type of AI machines are still not developed, but researchers are making lots of efforts and improvement for developing such AI machines.

Artificial Intelligence type-2: Based on functionality

4. Self-Awareness

- Self-awareness AI is the future of Artificial Intelligence. These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.
- These machines will be smarter than human mind.
- Self-Awareness AI does not exist in reality still and it is a hypothetical concept.

Application areas of AI

- AI in Healthcare
- AI in Astronomy
- AI in Gaming
- AI in Finance
- AI in Data Security
- AI in Social Media
- AI in Travel & Transport
- AI in Automotive Industry
- AI in Robotics
- AI in Entertainment
- AI in Agriculture
- AI in E-commerce
- AI in education

- Optical character recognition
- Handwriting recognition
- Speech recognition
- Face recognition
- Gaming
- Artificial creativity
- Computer vision, Virtual reality, and Image processing
- Photo and Video manipulation
- Diagnosis (artificial intelligence)

AI Problems:

AI Problems: also called as Task Domain in AI Applications

AI problems can be classified in three types:



Mundane Tasks

Are routine tasks / Common-place tasks

These tasks are done routinely by all people..

Mundane tasks include :

- Perception(Vision &Speech)
- NLP
- Common Sense reasoning
- Robotics
- Navigation

❖ Common-Place problems/tasks (Mundane Tasks):

- These are everyday tasks that humans and some animals do naturally and easily, but they are challenging for computers.

1. Recognizing People and Objects e.g. Your phone unlocks by recognizing your face.
2. Communicating Through Natural Language e.g. When you ask Siri or Alexa a question, and they respond.
3. Navigating Around Obstacles on the Streets e.g. Self-driving cars avoiding pedestrians and other vehicles.





❖ Formal problems/tasks (Structured Tasks):

- Formal tasks are structured problems that require logical thinking and precise algorithms.

1. Solving mathematical equations e.g.
   Statement: Solve $x+5=10$.
   Solution: Subtract 5 from both sides to get $x=5$.

2. Formal logic and theorem proving e.g.
   Statement: "If it rains, then the ground is wet." whose Logical Expression: $P \rightarrow Q$
   Statement: "The sum of the angles in a triangle is 180 degrees." which is based on: Euclidean geometry axioms

Formal Tasks

- Game Playing
- Mathematics/ Mathematical problem solving
- Logic
- Syllogism

Expert tasks:

- These tasks **cannot be done by all people, and can only be performed by skilled specialists.**
- *Clearly tasks of the first type are easy for humans to perform, and almost all are able to master them.*
- The second range of tasks requires **skill development and/or intelligence and only some specialists can perform them well.**
- However, when we look at what computer systems have been able to achieve to date, we see that their achievements include performing **sophisticated tasks like medical diagnosis, performing symbolic integration, proving theorems and playing chess.**

❖ Expert problems/Tasks:

- These are complex tasks that require specialized skills and knowledge, which only experts can do well.

1. Medical Diagnosis

   AI systems that help doctors by identifying diseases in medical images like X-rays.

2. Playing Games Like Chess

   Example: AI program Deep Blue that have beaten world champion in chess.





Expert Tasks include

- Engineering applications
- Scientific analysis
- Medical Analysis
- Financial Analysis


WHAT !
NEXT?

Outline

- Agent
- Agents and environments
- Rationality
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Environment types
- Agent types


AI Agents

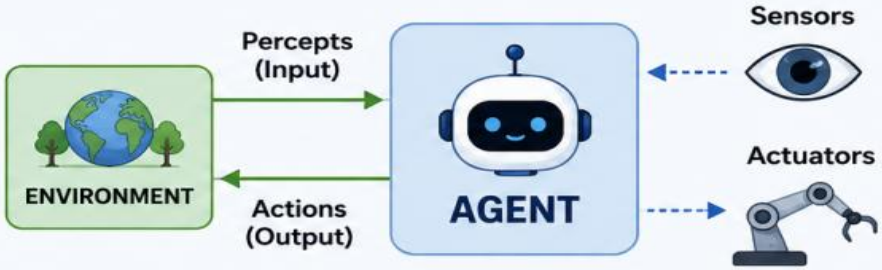


AGENT IN ARTIFICIAL INTELLIGENCE

DEFINITION

An agent is "anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators."

– Russell & Norvig



COMPONENTS OF AN AGENT

Sensors
Sensors are used by an agent to perceive and sense the environment.

Actuators
Actuators are used by an agent to take actions in the environment.

CHARACTERISTICS OF AN AGENT

- **Autonomy** – Operates without direct human intervention.
- **Reactivity** – Responds to changes in the environment.
- **Proactiveness** – Takes initiative to achieve goals.
- **Social Ability** – Can interact with other agents or humans.

EXAMPLES OF AGENTS



Smart Thermostat


Self-driving Car


Robot Vacuum Cleaner


Voice Assistant
(Alexa, Siri)



Email Spam
Filter


Game Playing
Agent



Stock Trading
Agent

Basic Architecture of Agent



- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.
  - **Human Sensors:**

    Eyes, ears, and other organs for sensors.
  - **Human Actuators:**

    Hands, legs, mouth, and other body parts.
  - **Robotic Sensors:**

    Mic, cameras and infrared range finders for sensors
  - **Robotic Actuators:**

    Motors, Display, speakers etc…

Agents :Introduction

An agent can be:

- **Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors

and hand, legs, vocal tract work for actuators.

- **Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for

sensors and various motors for actuators.

- **Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

Further

- Hence the world around us is full of agents such as thermostat, cell phone, camera, and even we are also agents. Before moving forward, following terms are important:
- **Sensor:** Sensor is a device which detects the change in the environment and sends the
- information to other electronic devices.
- An agent observes its environment through sensors.

Actuators: Actuators are the component of machines that **converts energy into motion**.

- The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

Effectors: Effectors are the devices ***which affect the environment.***

- Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

One very simple agent function is following: If the current square is dirty, then suck, otherwise move to other square.

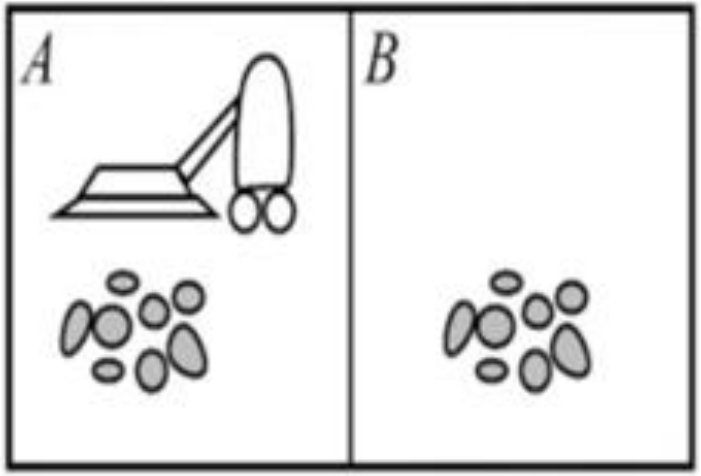
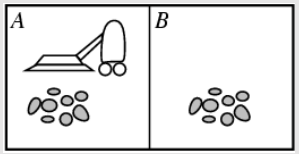


Figure (2) A vacuum cleaner world with just 2 locations.

Vacuum-cleaner world



Demo:
http://www.ai.sri.com/~oreilly/aima3ejava/aima3ejavademos.html

- Percepts: location and contents, e.g., [A,Dirty]
- Actions: *Left*, *Right*, *Suck*, *NoOp*
- *Agent's function → look-up table*
  - *For many agents this is a very large table*

Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Suck$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Suck$
$[A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Dirty]$	$Suck$
⋮	⋮

PEAS: Descriptor

- Performance Measure, Environment, Actuators, Sensors
- **Performance** :The output which get from the agent.
- All the necessary results that an agent gives after processing comes under its performance.
- **Environment:** All the surrounding things and conditions of an agent fall in this section. It basically consists of all the things under which the agents work.
- **Actuators :**The devices, hardware or software through which the agent performs any actions or processes any information to produce a result are the actuators of the agent.
- **Sensors:** The devices through which the agent observes and perceives its environment are the sensors of the agent

Example 1: The task of designing an **automated taxi driver.**

Examle 2: **Agent: Part-picking robot**

Automated taxi driver:

- Performance measure: Safe, fast, legal, comfortable trip, maximize profits
- Environment: Roads, other traffic, pedestrians, customers
- Actuators: Steering wheel, accelerator, brake, signal, horn
- Sensors: Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard

PEAS

- Agent: Part-picking robot
- Performance measure: Percentage of parts in correct bins
- Environment: Conveyor belt with parts, bins
- Actuators: Jointed arm and hand
- Sensors: Camera, joint angle sensors

Environment(as per AI Terminology)

- An environment can **be described as a situation in which an agent is present/or will perform…**

- The environment is **where agent lives, operate and provide the agent with something to sense and act upon it.**

Properties of Environment

1) Fully observable
2) Partially Observable
3) Static
4) Dynamic
5) Discrete
6)Continuous
7) Deterministic
8) Stochastic
9) Single-agent
10) Multi-agent
11) Episodic
12) Sequential
13) Known
14) Unknown

1.Fully observable vs Partially Observable:

- If an agent sensor can sense or access the **complete state of an environment** at each point of time then it is a fully observable environment, else ***it is partially observable.***
- A fully observable environment is easy as there is no need to maintain the internal state to keep track history of the world.
- Enables better decisions..
- Partially observable is incomplete..

- Note :An agent with no sensors in all environments then such an environment is called as unobservable.



Example: chess – the board is fully observable
For Partially ???

Environment

Examples :

- **Real-life Example**: While running a car on the road (**Environment**), The driver (**Agent**) is able to see road conditions, signboard and pedestrians on the road at a given time and drive accordingly.
  So Road is a **fully observable environment** for a driver while driving the car.

- **Real-life Example**: Playing card games is a perfect example of a **partially-observable environment** where a player is not aware of the card in the opponent's hand. Why partially-observable? Because the other parts of the environment, e.g opponent, game name, etc are known for the player (Agent).

2. Deterministic vs Stochastic

- If an agent's current state and selected action ***can completely determine the next state of the environment,*** then such environment is called a **deterministic environment.**

- A ***stochastic environment is random in nature*** and cannot be determined completely by an agent.
- In a deterministic, fully observable environment, agent does not need to worry about uncertainty

Example: Cross world Puzzle
Taxi driving(stochastic)

Environment

Examples :

- **Real-life Example**: The traffic signal is a deterministic environment where the next signal is known for a pedestrian (**Agent**)

- **Real-life Example:** The radio station is a stochastic environment where the listener is not aware about the next song or playing a soccer is stochastic environment

3.Episodic vs Sequential

- In an episodic environment, **there is a series of one-shot actions,** and only the current percept is required for the action.

In this agent's experience is divided into atomic episodes which are independent ….

- However, in Sequential environment, ***an agent requires memory of past actions*** to determine the next best actions.

Environment

But some times ,its not possible to achieve exactly the same

- **Episodic** is an environment where each state is independent of each other. The action on a state has nothing to do with the next state.
- **Real-life Example**: A support bot (agent) answer to a question and then answer to another question and so on. So each question-answer is a single episode.
- **The sequential** environment is an environment where the next state is dependent on the current action. So agent current action can change all of the future states of the environment.
- **Real-life Example**: Playing tennis is a perfect example where a player observes the opponent's shot and takes action.

4. Static vs Dynamic

- ***If the environment can change itself*** while an agent is deliberating then such environment is called a dynamic environment else it is called a static environment.
- ***Static environments are easy to deal*** because an agent does not need to continue looking at the world while deciding for an action.
- However for dynamic environment, agents need to keep looking at the world at each action.
- ***Taxi driving is an example of a dynamic environment*** whereas Crossword puzzles are an example of a static environment.

Environment

Examples :

- The **Static** environment is completely unchanged while an agent is precepting the environment.
- **Real-life Example**: Cleaning a room (Environment) by a dry-cleaner reboot (Agent) is an example of a static environment where the room is static while cleaning.
- **Dynamic** Environment could be changed while an agent is precepting the environment. So agents keep looking at the environment while taking action.
- **Real-life Example**: Playing soccer is a dynamic environment where players' positions keep changing throughout the game. So a player hit the ball by observing the opposite team.

5. Discrete vs Continuous

- If in an environment there are **a finite number of precepts and actions** that can be performed within it, then such an environment is called a **discrete environment** else it is called continuous environment.
- Continuous environment **sweeps through a range of values..**
- A chess game comes under discrete environment as there is a finite number of moves that can be performed.
- A self-driving car is an example of a continuous environment.

Environment

Discrete / Continuous

Discrete

- The environment consists of a finite number of actions that can be deliberated in the environment to obtain the output.
- **Example**: chess



Continuous

- The environment in which the actions performed cannot be numbered ie.,is not discrete, is said to be continuous.
- **Example**: Self-driving cars



Examples :

- **Real-life Example**: Choices of a move (action) in a tic-tac game are finite on a finite number of boxes on the board (Environment).
- **Real-life Example**: In a basketball game, the position of players (**Environment**) keeps changing continuously and hitting (**Action**) the ball towards the basket can have different angles and speed so infinite possibilities.

6. Single-agent vs Multi-agent

- ***If only one agent is involved*** in an environment, and operating by itself then such an environment is called single agent environment.

- However, **if multiple agents are operating** in an environment, then such an environment ***is called a multi-agent environment.***

- The agent design problems in the multi-agent environment are different from single agent environment

Example : Cross word puzzle..
And for Multi agent???

Environment

Single-agent / Multi-agent

Single-agent

- If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
- **Example**: maze



Multi-agent

- if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.
- **Example**: football



Examples:

- **Real-life Example**: Playing tennis against the ball is a single agent environment where there is only one player.

- **Real-life Example**: Playing a soccer match is a multi-agent environment.

Few of the examples

Types of Agent

Basic Architecture of Agent

Structure of an AI Agent

- The task of AI is to design an agent program which implements the agent function
- Agent is a combination of architecture and agent program

Agent = Architecture + Agent program

- **Architecture:** Architecture is machinery that an AI agent executes on.
- **Agent Function:** Agent function is used to map a percept to an action.

$$f:P^* \rightarrow A$$

- **Agent program:** Agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function (f).

Rational Agent -

- A system is rational if it does the “right thing”. Given what it knows.

Characteristic of Rational Agent :

- The agent's prior knowledge of the environment.
- The performance measure that defines the criterion of success.
- The actions that the agent can perform.
- The agent's percept sequence to date.

- For every possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Rationality

- Rationality is nothing but status of being reasonable, sensible, and having good sense of judgment.
- Rationality is concerned with expected actions and results depending upon what the agent has perceived.
- Performing actions with the aim of obtaining useful information is an important part of rationality.
- A rational agent always performs right action, where the right action means the action that causes the agent to be most successful in the given percept sequence

TYPES OF AGENTS

- Agents can be grouped into four classes **based on their degree of perceived intelligence and capability :**
- Simple Reflex Agents
- Model-Based Reflex Agents
- Goal-Based Agents
- Utility-Based Agents
- Learning Agent

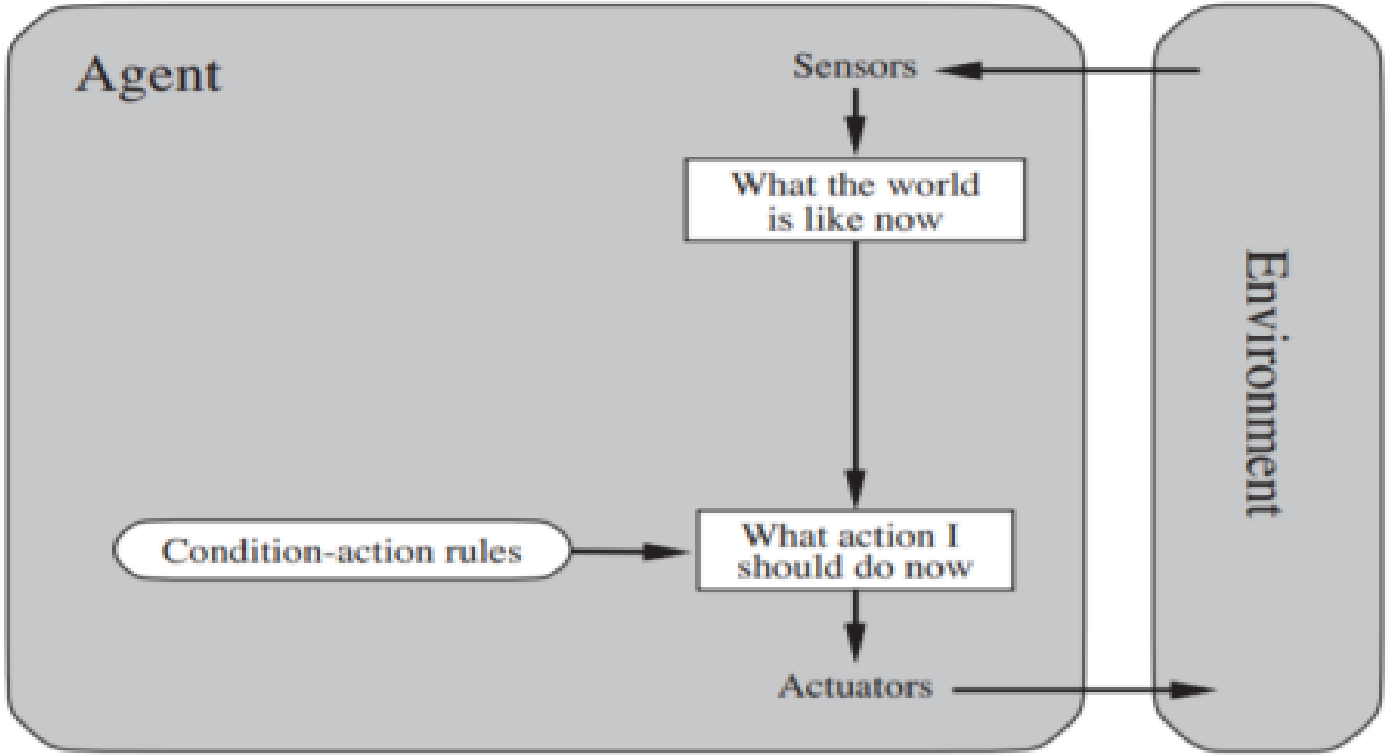


Figure 1.6 A simple reflex agent

The Simple reflex agents

- The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percepts and ignore the rest of the percept history (past State).
- These agents **only succeed in the fully observable environment.**
- The Simple reflex agent does not consider any part of percepts history during their decision and action process.
- The Simple reflex agent **works on Condition-action rule,** which means it maps the current state to action.
- Such as a ***Room Cleaner agent,*** it works only if there is dirt in the room.

Mostly too big to generate and to store.

- Not adaptive to changes in the environment.
- **Condition-Action Rule** – It is a rule that maps a state (condition) to an action. Ex: if car-in-front-is-braking then initiate- braking.

Simple reflex agents

```
def SIMPLE-REFLEX-AGENT( percept)
persistent: rules, a set of condition-action rules

    state ← INTERPRET-INPUT(percept)
    rule ← RULE-MATCH(state, rules)
    action ← rule.ACTION
    return action
```

Problems for the simple reflex agent design approach:

- They have very limited intelligence
- They do not have knowledge of non-perceptual parts of the current state

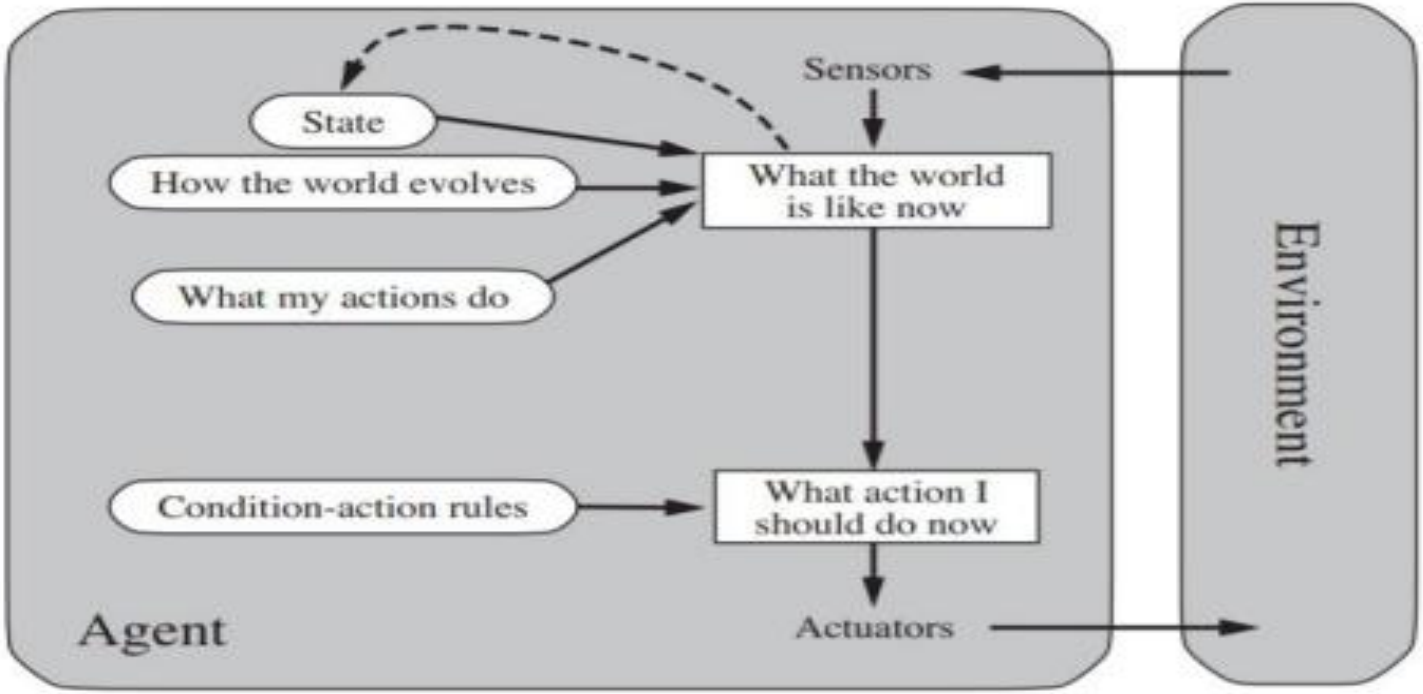


Figure 1.7 A model-based reflex agent

Model Based Reflex Agents

- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
- Model: It is knowledge about "how things happen in the world," so it is called a Model-based agent.
- Internal State: It is a representation of the current state based on percept history.
- These agents have the model, **"which is knowledge of the world"** and based on the model they perform actions.
- Updating the agent state requires information about:
  - How the world evolves
  - How the agent's action affects the world.

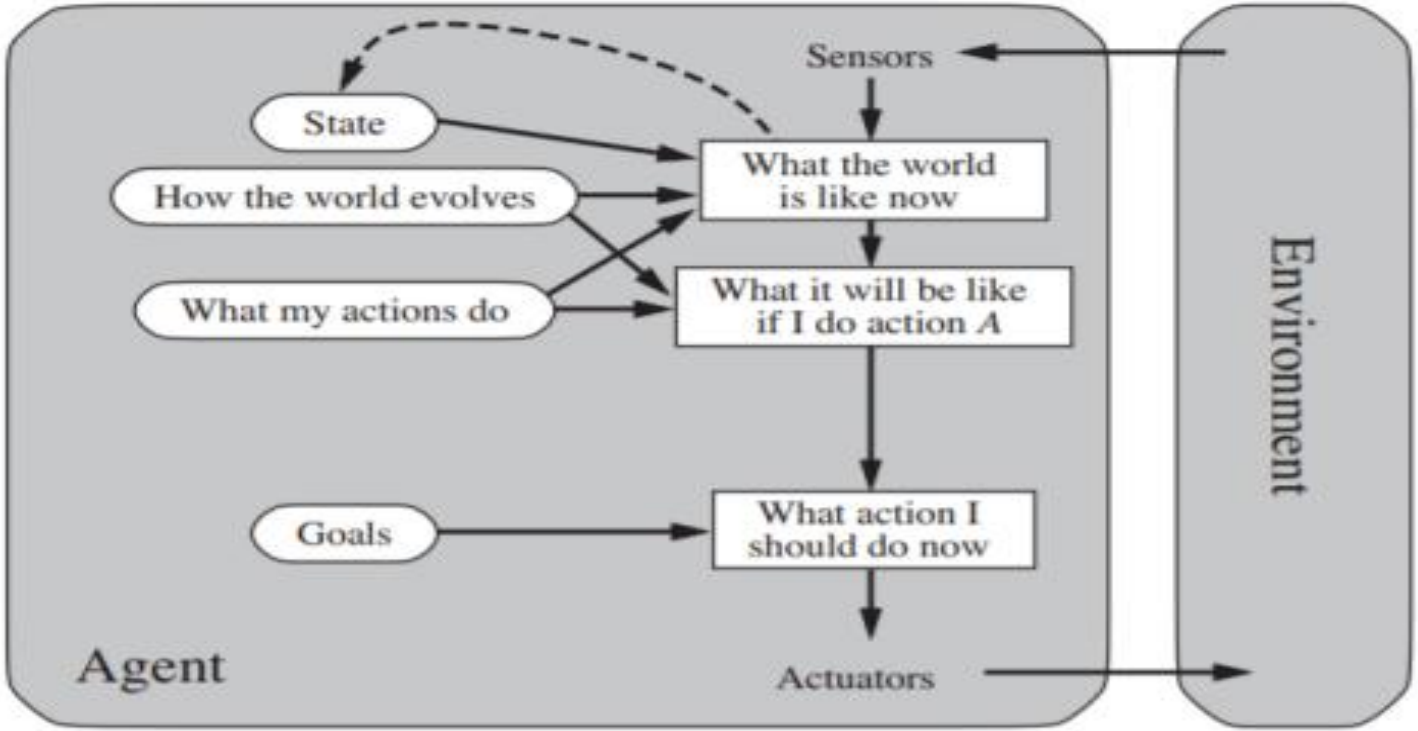


Figure 1.8 A goal-based agent

Goal Based Agents

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to **know its goal which describes desirable situations**.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- **They choose an action, so that they can achieve the goal.**
- These agents may **have to consider a long sequence of possible actions** before deciding whether the goal is achieved or not.
- Such considerations of different scenario **are called searching and planning, which makes an agent proactive**.

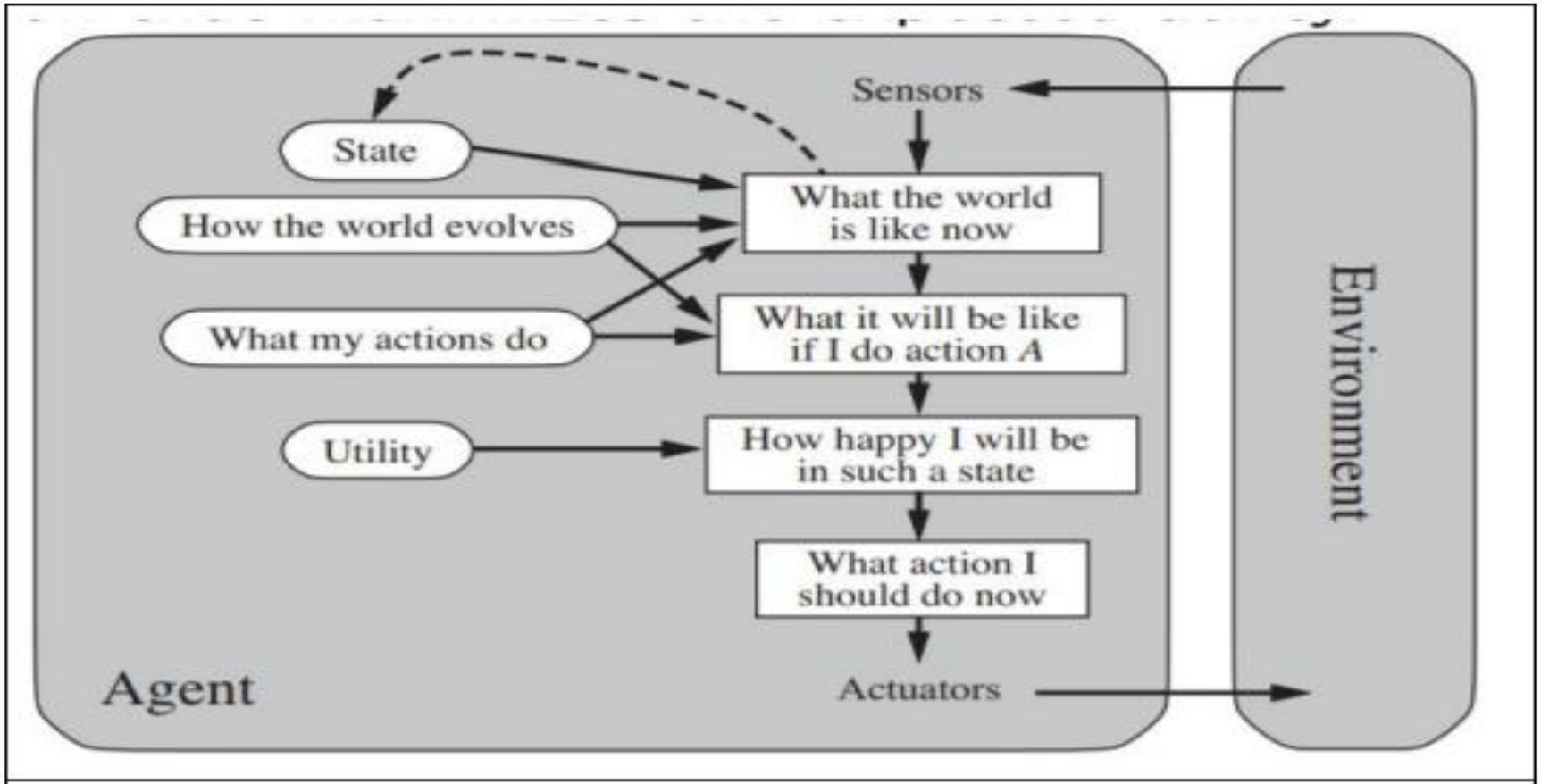


Figure 1.9 A utility-based agent

Utility Based Agents

- These agents are similar to the goal-based agent but provide an extra component of utility measurement **("Level of Happiness")** which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals **but also the best way to achieve the goal.**
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.

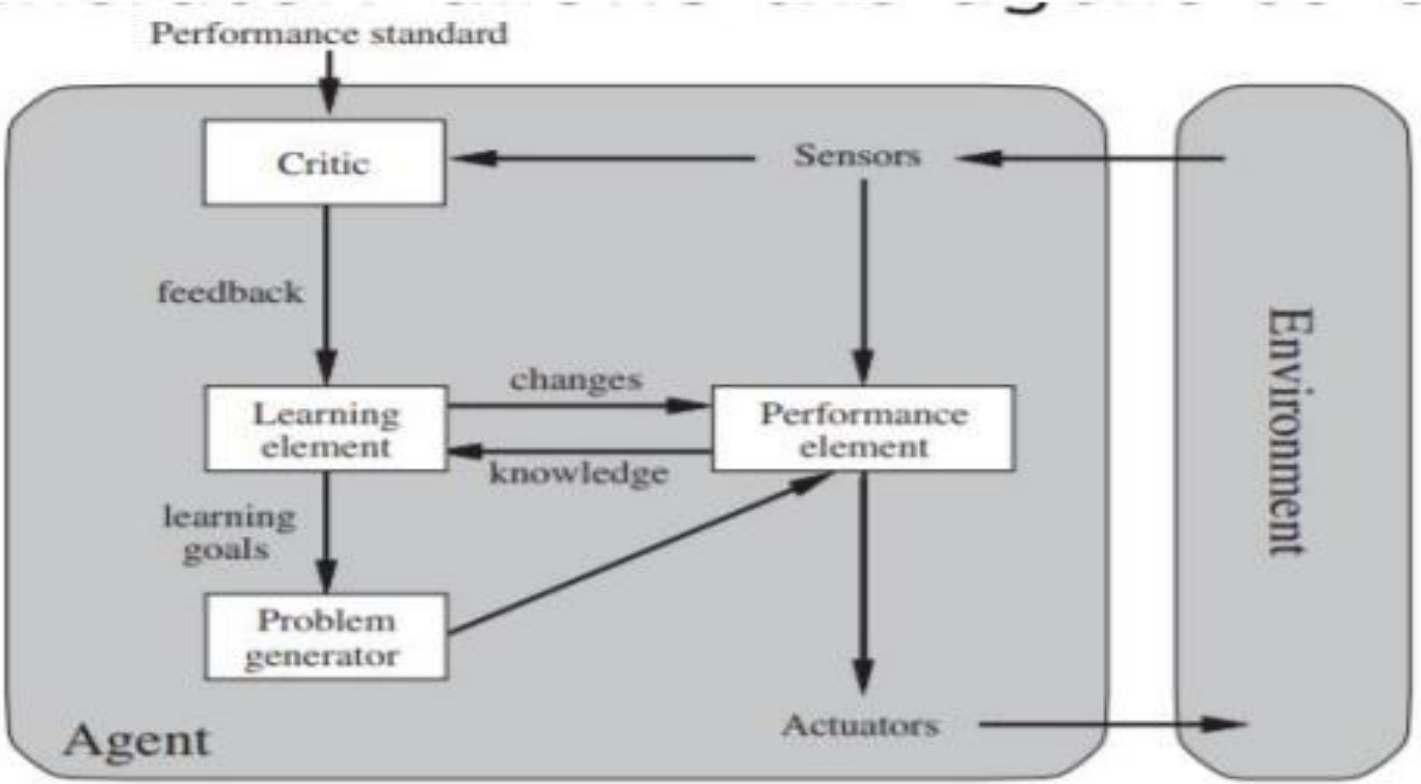


Figure 1.10 Learning Agents

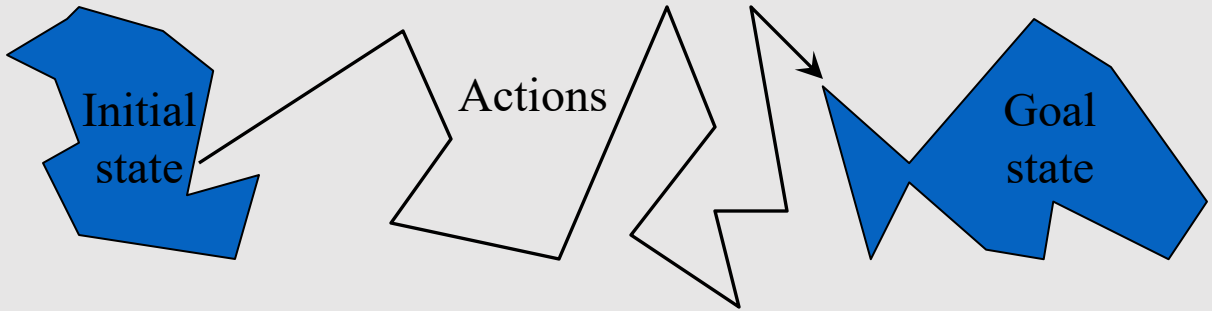
Learning Agents

- A learning agent in AI is the type of agent **which can learn from its past experiences, or it has learning capabilities.**
- It starts to act with basic knowledge and then able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
- Learning element: It is responsible for making improvements by learning from environment
- Critic: Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard. c. Performance element: It is responsible for selecting external action
- Problem generator: This component is responsible for suggesting actions that will lead to new and informative experiences.
- Hence, learning agents are **able to learn, analyze performance, and look for new ways to improve the performance.**

Building goal-based agents

To build a goal-based agent we need to answer the following questions:

- What is the goal to be achieved?
- What are the actions?
- What is the representation?
  - E.g., what *relevant* information is necessary to encode in order to describe the state of the world, describe the available transitions, and solve the problem?)



PROBLEM SOLVING APPROACH TO TYPICAL AI PROBLEMS

- Problem formulation: is a special kind of goal based agent
- PSA decides what to do
- Find sequence of actions to solve it

PSA has the following 3 key words:

1. FORMULATE
2. SEARCH
3. EXECUTE

PROBLEM FORMULATION IS DESCRIBED BY :

- INITIAL STATE
- SUCCESSOR FUNCTION
- STATE SPACE
- PATH SPACE
- GOAL TEST
- PATH COST
- STEP COST

GOAL FORMULATION HAS THE FOLLOWING

- GOAL COSTS:
- The costs of the final path selected
- PATH COSTS:
- The cost given to each path

Evaluation criteria:

- **completeness**: will it find a solution if one exists?
- **time efficiency**:
- **space efficiency**:
- **optimality of solution**:

1. Example Problems of PSA:
    - Toy problems:
        1. 8-puzzle
        2. 8-queen/n-queen
        3. cryptarithmetic
        4. vacuum world
        5. missionaries and cannibals
    - Real World
        1. Traveling Salesperson (NP hard)
        2. VLSI layout
        3. robot navigation
        4. assembly sequencing

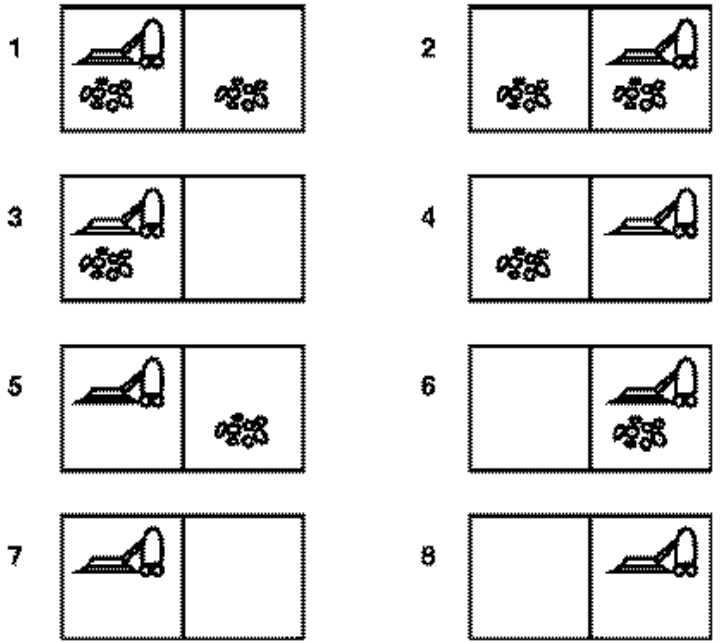
Vacuum cleaner:

- States: 8
- Initial state: any idle state
- Successor function:

  left,right suck,idle
- Goal Test: clean the surface
- Path Cost: each step:1 (step may be towards left/right/suck)

1. **Single state problems**

Consider the **vacuum cleaner world.**



Imagine that our intelligent agent is a robot vacuum cleaner.

- States: 8
- Initial state: any idle state
- Successor function:

  left,right suck,idle
- Goal Test: clean the surface
- Path Cost: each step:1 (step may be towards left/right/suck)

8-PUZZLE

- States:
- Initial state:
- Successor function:

- Goal Test:

- Path Cost:

- States: 9!(3x3)(rows/columns)
- Initial state: any idle state
- Successor function:

  left,right up,down
- Goal Test: sequence from 1-8(in order)
- Path Cost: each step:1

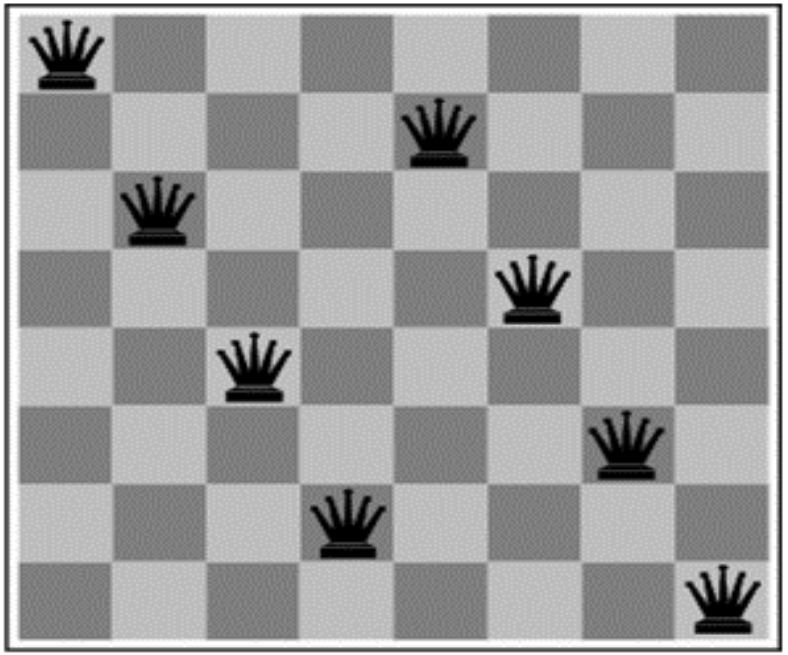
The 8-Queens Problem

State Representation: ?

Initial State: ?

Operators: ?

Goal: Place eight queens on a **chessboard such that no queen attacks any other!**



8-Queen Problem

Initial States: any arrangement of 0 to 8 queens on the board.

Operators: place a queen in the left-most empty column such that it is not attacked by any other queen.

The objective is to place eight queens on a chessboard such that no queen threatens another.

A queen threatens another queen in the same row, column or diagonal.

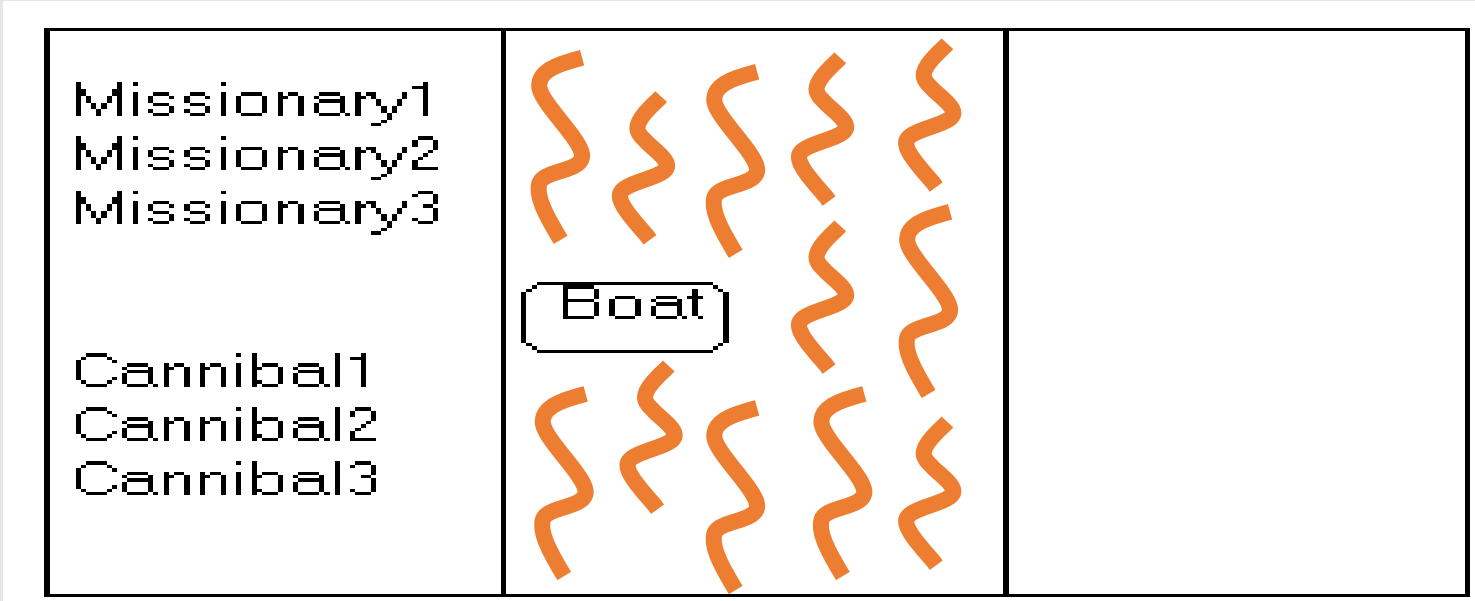
2057 possible sequences to investigate.

Representing states

- What information is necessary to encode about the world to sufficiently describe all relevant aspects to solving the goal?
- That is, what knowledge needs to be represented in a state description to adequately describe the current state or situation of the world?
- The **size of a problem** is usually described in terms of the **number of states** that are possible.
  - The 8-puzzle has **181,440 states.**
  - Tic-Tac-Toe has about **3^9 states.**
  - Rubik's Cube has about **10^{19} states.**
  - Checkers has about **10^{40} states**.
  - Chess has about **10^{120} states in a typical game.**

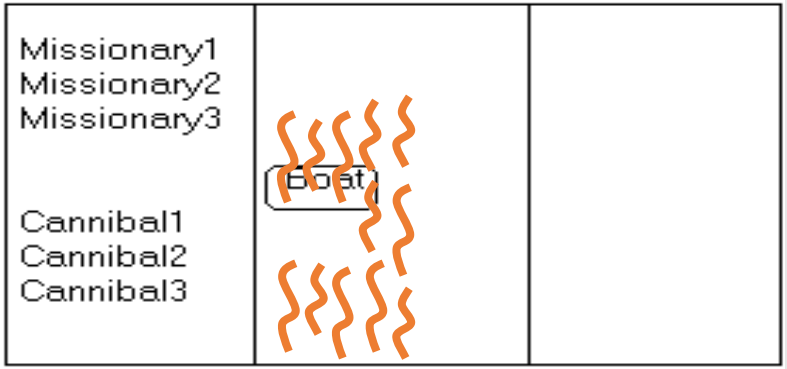
Missionaries and Cannibals

Three missionaries and three cannibals wish to cross the river. They have a small boat that will carry up to two people. Everyone can navigate the boat. If at any time the Cannibals outnumber the Missionaries on either bank of the river, they will eat the Missionaries. Find the smallest number of crossings that will allow everyone to cross the river safely.


Missionary1
Missionary2
Missionary3
Boat
Cannibal1
Cannibal2
Cannibal3

Missionaries and Cannibals

- **Goal**: Move all the missionaries and cannibals across the river.
- **Constraint:** Missionaries can never be outnumbered by cannibals on either side of river, or else the missionaries are killed.
- **State:** configuration of missionaries and cannibals and boat on each side of river.
- **Initial State:** 3 missionaries, 3 cannibals and the boat are on the near bank
- **Operators:** Move boat containing some set of occupants across the river (in either direction) to the other side.



3 Missionaries and 3 Cannibals wish to cross the river. They have a boat that will carry two people. Everyone can navigatethe boat. If at any time the Cannibals outnumber the missionaries on either bank of the river, they will eat the Missionaries. Find the smallest number of crossings that will allow everyone to cross the river safely.
The problem can be solved in 11 moves. But people rarely get the optimal solution, because the MC problem contains a 'tricky' state at the end, where two people move back across the river.

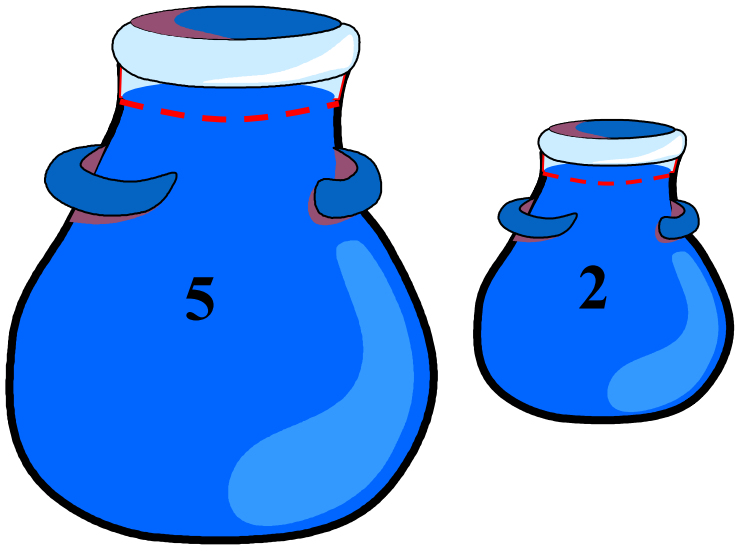
Missionaries and Cannibals Solution

	Near side	***Far side***
0 Initial setup:	MMMCCC B	-
1 Two cannibals cross over:	MMMC B	CC
2 One comes back:	MMMCC B	C
3 Two cannibals go over again:	MMM B	CCC
4 One comes back:	MMMC B	CC
5 Two missionaries cross:	MC B	MMCC
6 A missionary & cannibal return:	MMCC B	MC
7 Two missionaries cross again:	CC B	MMMC
8 A cannibal returns:	CCC B	MMM
9 Two cannibals cross:	C B	MMMCC
10 One returns:	CC B	MMMC
11 And brings over the third:	- B	MMMCCC

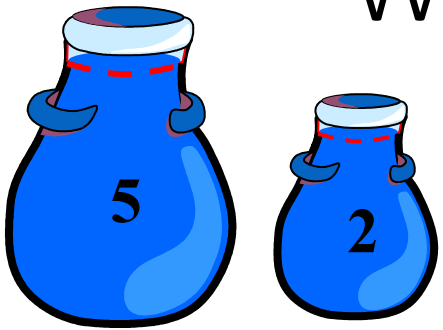
Water Jug Problem

Given a full 5-gallon jug and a full 2-gallon jug, fill the 2-gallon jug with exactly one gallon of water.

- **State:** ?
- **Initial State:** ?
- **Operators:** ?
- **Goal State:** ?



Water Jug Problem



- State = (x,y), where x is the number of gallons of water in the 5-gallon jug and y is # of gallons in the 2-gallon jug
- Initial State = (5,2)
- Goal State = (*,1), where * means any amount

Operator table

Name	Cond.	Transition	Effect
Empty5	–	(x,y)→(0,y)	Empty 5-gal. jug
Empty2	–	(x,y)→(x,0)	Empty 2-gal. jug
2to5	$x \leq 3$	(x,2)→(x+2,0)	Pour 2-gal. into 5-gal.
5to2	$x \geq 2$	(x,0)→(x-2,2)	Pour 5-gal. into 2-gal.
5to2part	$y < 2$	(1,y)→(0,y+1)	Pour partial 5-gal. into 2-gal.